

NAME:

M534H Take Home Final Exam

Due on: Tuesday May 12, 2026 no later than at 5:30pm

Instructions.

- (1) This exam consists of 4 problems with parts for a total of 100%.
- (2) Type in Latex each problem and its solution in an ordered fashion (new page for each problem)
- (3) Show all the work needed to reach your answer for full credit.
- (4) You should work on it alone. You may consult W. Strauss' book, the class notes and your homework **only**. No other material is allowed. No use of LLM/web searchers or inquiries is allowed
- (5) You cannot discuss the problems with other people, including your classmates. You cannot do any web searches or inquiries
- (6) Upload your final exam (PDF document) to Canvas. Contact me via email asap if you encounter any issues with Canvas.

Problem 1.

Consider the *initial value problem for the wave equation* on $\mathbb{R}$:

$$\begin{cases} u_{tt} - u_{xx} = 0 \\ u(x, 0) = \phi(x) \\ u_t(x, 0) = \psi(x) \end{cases}$$

where $\phi, \psi : \mathbb{R} \rightarrow \mathbb{R}$ are two smooth given functions (data). Let $x_0 \in \mathbb{R}$ $t_0 > 0$ be fixed and suppose that $\phi(x)$ and $\psi(x)$ vanish for all x in the interval $[x_0 - t_0, x_0 + t_0]$.

Finite Propagation Speed Theorem.: The solution $u(x, t)$ to the initial value problem above vanishes for all (x, t) within $\mathcal{C}$, the domain of dependence of (x_0, t_0) . Recall

$$\mathcal{C} := \{(x, t) : 0 \leq t \leq t_0 \text{ and } x_0 - (t_0 - t) \leq x \leq x_0 + (t_0 - t)\}.$$

Remark: The Theorem above is also valid in higher dimensions but for simplicity above we consider it only in one (space) dimension. In one dimension, one can trivially prove the above theorem directly using the representation formulas for the solution $u(x, t)$ in terms of the initial data which are available in one dimension. Or, one could prove it without using this explicit representation of u , but by using the *energy method* instead –as we have seen in class-. This proof is a bit harder but the advantage of the method is that it also works in higher dimensions. This Problem is then to **prove the Finite Propagation Speed Theorem using the energy method. You must follow the steps below:**

First for each $0 \leq t \leq t_0$, let $I_t := [x_0 - (t_0 - t), x_0 + (t_0 - t)]$. Note I_t is contained in the interval $(x_0 - t_0, x_0 + t_0)$.

Next define the modified energy:

$$\tilde{E}(t) = \frac{1}{2} \int_{I_t} |u_t|^2 + |u_x|^2 dx$$

Note that $\tilde{E}(t) \geq 0$ for any t and that $\mathcal{C} = \bigcup_{0 \leq t \leq t_0} I_t$.

The goal is to show that for each $0 \leq t \leq t_0$, $u(x, t) = 0$ for all $x \in I_t$. Do so by proving the following:

(1) Prove that $\tilde{E}(t)$ is a decreasing function of t by showing that $\frac{d\tilde{E}}{dt} \leq 0$

To compute the derivative in time use: (see A.3 Theorem 3 in Strauss's book p.421).

$$\frac{d}{dt} \int_{a(t)}^{b(t)} F(x, t) dx = \int_{a(t)}^{b(t)} \frac{d}{dt} F(x, t) dx + [F(b(t), t)b'(t) - F(a(t), t)a'(t)]$$

(2) Show that $\tilde{E}(0) = 0$

(3) By (1) you then have that $\tilde{E}(t) \leq \tilde{E}(0)$ for any $0 \leq t \leq t_0$ and by (1) you can conclude that $\tilde{E}(t) = 0$ for any $0 \leq t \leq t_0$. Prove then that this implies that $u(x, t) = 0$ for any $x \in I_t$ and any $0 \leq t \leq t_0$.

Problem 2

a) Consider the initial value problem for the diffusion equation on the whole real line $\mathbb{R}$ with $k = 1$:

$$(*) \quad \begin{cases} u_t - u_{xx} = 0, & x \in \mathbb{R}, \ t > 0 \\ u(x, 0) = e^{2x}, & x \in \mathbb{R} \end{cases}$$

Use the fact that the solution $u(x, t)$ is obtained by the convolution of the fundamental solution with the initial data; that is, use that:

$$u(x, t) = \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{4kt}} e^{2y} dy$$

to find the function that $u(x, t)$ equals to. You must follow the following steps:

i) Develop the square in $e^{-\frac{(x-y)^2}{4kt}}$, and collect all the exponents of the exponentials and **complete the square in the y variable**. Note that terms that have only x and t in the exponents can come out of the integral.

ii) Change of variables by letting $p := \frac{y-(x+4t)}{\sqrt{4t}}$. Use that $\int_{-\infty}^{\infty} e^{-p^2} dp = \sqrt{\pi}$.

b) Verify that the solution $u(x, t)$ you found in a) satisfies the equation $(*)$.

Problem 3

Let $u_1(x, t)$ and $u_2(x, t)$ be solutions to the heat equation $u_t = k u_{xx}$ with initial and boundary conditions:

$$u_1(x, 0) = f_1(x), \quad u_1(0, t) = g_1(x), \quad u_1(L, t) = h_1(t)$$

and

$$u_2(x, 0) = f_2(x), \quad u_2(0, t) = g_2(x), \quad u_2(L, t) = h_2(t)$$

respectively.

Assume that $f_1 \geq f_2$, $g_1 \geq g_2$ and that $h_1 \geq h_2$.

Prove then that $u_1 \geq u_2$ in the region $\mathcal{R} = [0, L] \times [0, \infty)$.

Hint: Consider $w = u_1 - u_2$, set up an appropriate boundary-initial value problem for w and use the (weak) max or the min principle (specify) to prove that $w \geq 0$ on $\mathcal{R}$.

Problem 4

a) Find the Fourier cosine series of $\phi(x) = x^2$ for $x \in [0, 1]$

b) Use separation of variables and the superposition principle to find the general solution to the following boundary value problem for the heat equation on an interval:

$$(H) \quad \begin{cases} u_t - u_{xx} = 0, & 0 < x < 1, \ t > 0 \\ u_x(0, t) = 0 = u_x(1, t) & t > 0 \end{cases}$$

using separation of variables, that is set $u(x, t) = X(x)T(t)$ and fully solve the problem of finding $X(x)$ and $T(t)$.

In the course of your proof justify do an analysis of all the possible eigenvalues ($\lambda > 0$, $\lambda = 0$, $\lambda < 0$) to the problem,

$$\begin{cases} X'' + \lambda X(x) = 0 & 0 < x < 1 \\ X'(0) = 0 = X'(1) \end{cases}$$

c) Find the particular solution to (H) that also satisfies the initial condition that $u(x, 0) = x^2$, for $0 < x < 1$.